

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	139782	46.022856
1	[Git, Change, Create]	45409	14.950794
2	[Git, Change, Delete]	22797	7.505852
3	[Java, ByteCode, Member, Method]	13783	4.538017
4	[Java, ByteCode, Parameter]	13447	4.427389
5	[Git, Commit]	11584	3.814002
6	[File, Git]	8331	2.742960
7	[Java, ByteCode, Bound, ParameterizedType]	7316	2.408774
8	[Java, ByteCode, Bound]	7304	2.404823
9	[Git, Change, Rename]	7018	2.310658
10	[Java, ByteCode, Member, Field]	3704	1.219532
11	[Java, Value, ByteCode, Annotation]	2955	0.972926
12	[Java, ByteCode, Bound, WildcardType]	2950	0.971280
13	[Xml, Element]	2162	0.711833
14	[Java, ByteCode, Member, Method, Constructor]	2159	0.710845
15	[Xml, Text]	1450	0.477409
16	[Java, ByteCode, TypeVariable, Bound]	1112	0.366123
17	[Java, ByteCode, Member, Method, Lambda]	986	0.324638
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	903	0.297310
19	[Type, File, Java, ByteCode, Class]	868	0.285787
20	[Json, Key]	739	0.243314
21	[Value, Json, Scalar]	723	0.238046
22	[Java, Value, ByteCode, Primitive]	685	0.225534
23	[Type, File, Java, ByteCode, JavaType]	659	0.216974
24	[Java, ByteCode, GenericDeclaration, Member, M...	579	0.190634
25	[Type, File, Java, ByteCode, ExternalType]	419	0.137955
26	[Git, Change, Copy]	298	0.098116
27	[Author, Git, Person]	297	0.097786
28	[Value, Array]	293	0.096469
29	[Committer, Git, Person]	242	0.079678

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

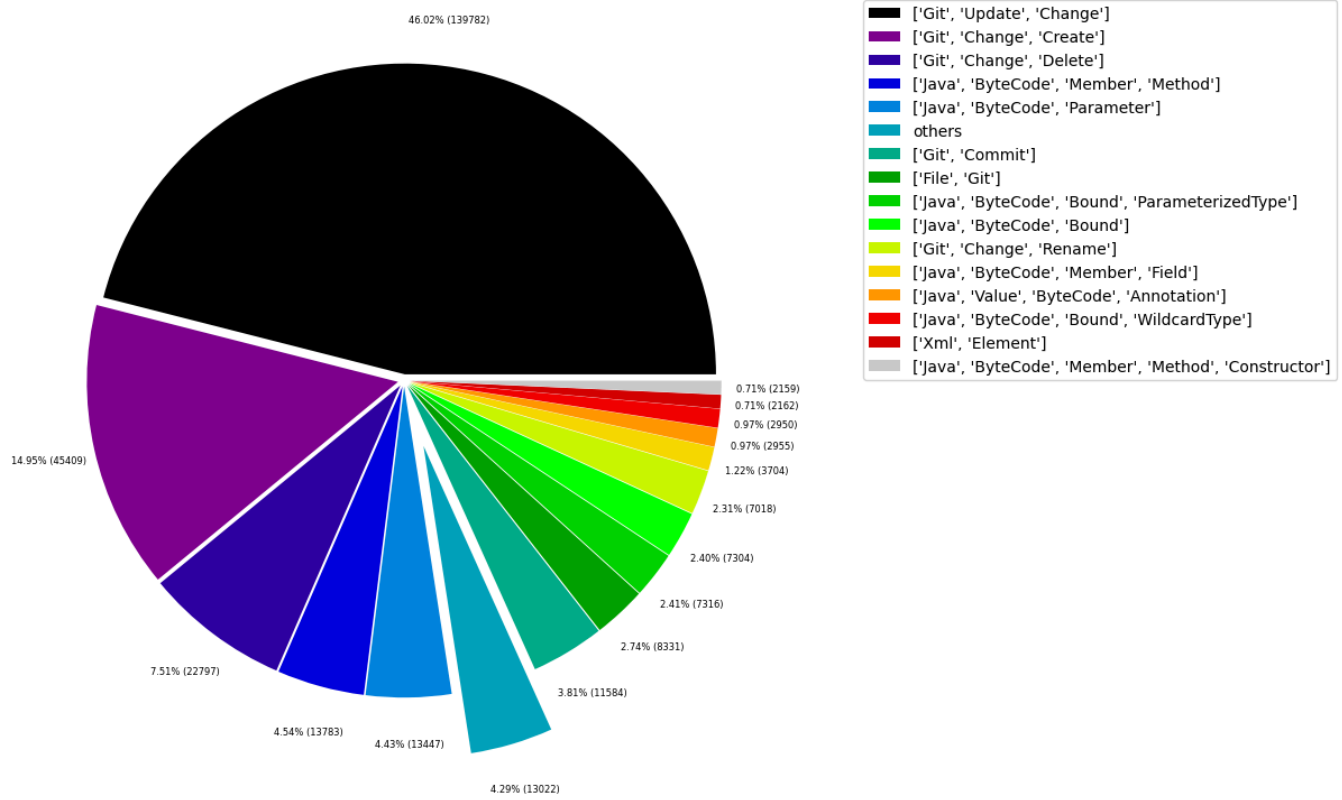


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000329
1	[Git, Branch]	1	0.000329
2	[Repository, File, Git]	1	0.000329
3	[File, Json]	2	0.000658
4	[File]	3	0.000988
5	[Java, ByteCode, GenericDeclaration, Member, M...	4	0.001317
6	[Maven, Exclusion]	5	0.001646
7	[Value, Array, Json]	6	0.001975
8	[Type, File, Java, ByteCode, Void]	9	0.002963
9	[File, Maven, Xml, Pom, Document]	9	0.002963
10	[Java, ManifestSection]	9	0.002963
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002963
12	[File, Java, Manifest]	9	0.002963
13	[File, Java, ServiceLoader]	11	0.003622
14	[File, Java, Properties]	12	0.003951
15	[Maven, ExecutionGoal]	16	0.005268
16	[Maven, PluginExecution]	16	0.005268
17	[Xml, Attribute]	18	0.005926
18	[jQAssistant, Rule, Concept]	19	0.006256
19	[Maven, Plugin]	21	0.006914
20	[Maven, Configuration]	21	0.006914
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006914
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007243
23	[Type, File, Java, ByteCode, Enum]	30	0.009877
24	[Type, File, Java, ByteCode, PrimitiveType]	31	0.010207
25	[Xml, Namespace]	36	0.011853
26	[Type, File, Java, ByteCode, Annotation]	44	0.014487
27	[File, Directory]	45	0.014816
28	[Type, File, Java, ByteCode, Class, Throwable]	56	0.018438
29	[Type, File, Java, ByteCode, GenericDeclaratio...	85	0.027986

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

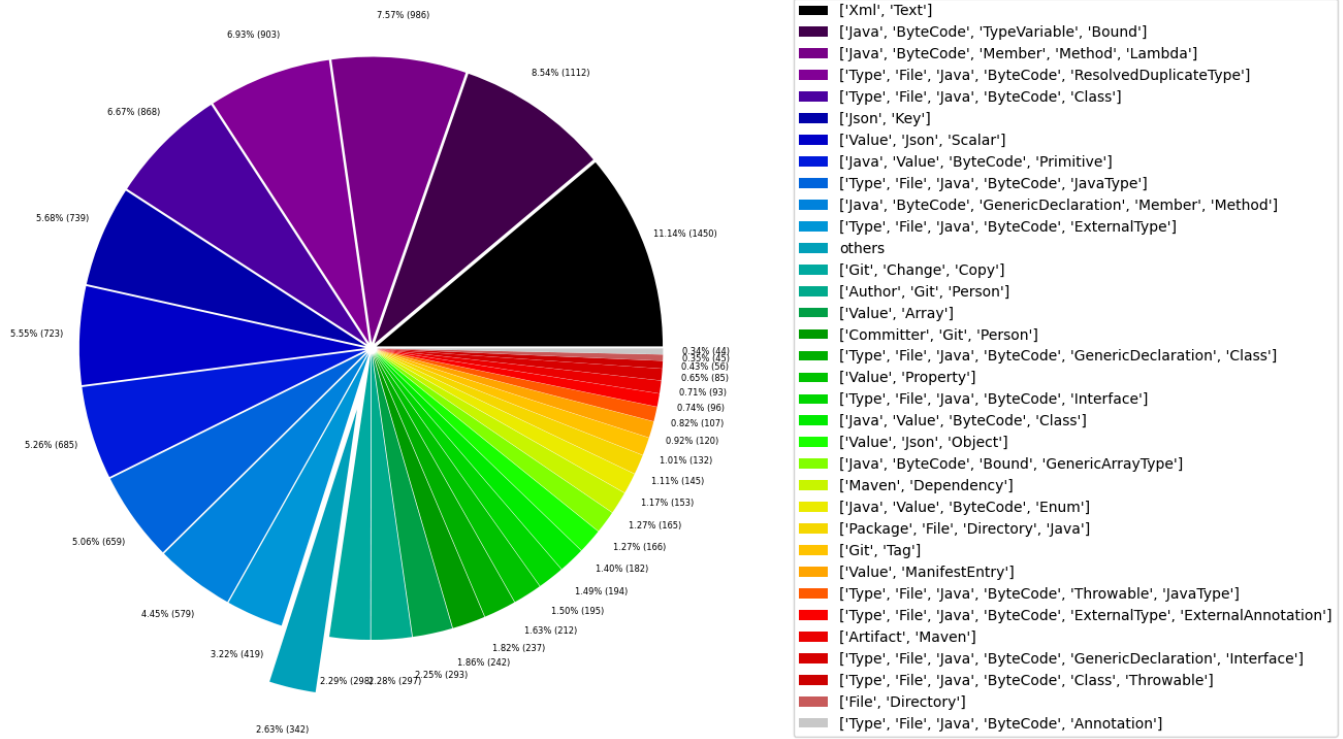


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

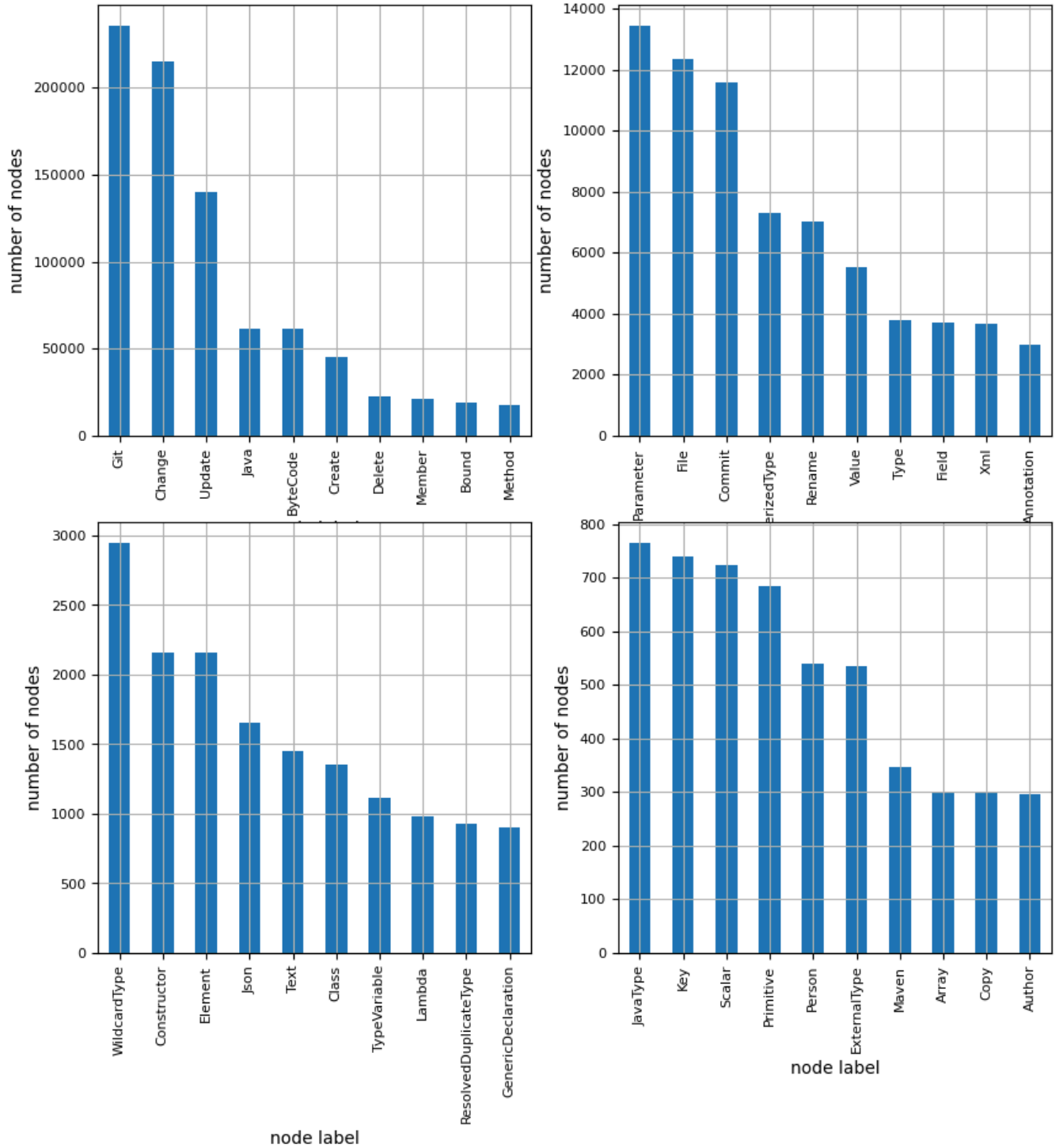
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	235892	77.666821
1	Change	215304	70.888276
2	Update	139782	46.022856
3	Java	61474	20.240153
4	ByteCode	61279	20.175950
5	Create	45409	14.950794
6	Delete	22797	7.505852
7	Member	21215	6.984983
8	Bound	18848	6.205654
9	Method	17511	5.765451
10	Parameter	13447	4.427389
11	File	12359	4.069168
12	Commit	11584	3.814002
13	ParameterizedType	7316	2.408774
14	Rename	7018	2.310658
15	Value	5523	1.818433
16	Type	3782	1.245214
17	Field	3704	1.219532
18	Xml	3675	1.209984
19	Annotation	2999	0.987413
20	WildcardType	2950	0.971280
21	Constructor	2163	0.712162
22	Element	2162	0.711833
23	Json	1652	0.543917
24	Text	1450	0.477409
25	Class	1355	0.446130
26	TypeVariable	1112	0.366123
27	Lambda	986	0.324638
28	ResolvedDuplicateType	925	0.304554
29	GenericDeclaration	905	0.297969
30	JavaType	766	0.252203
31	Key	739	0.243314
32	Scalar	723	0.238046
33	Primitive	685	0.225534
34	Person	539	0.177464
35	ExternalType	536	0.176477
36	Maven	346	0.113920
37	Array	299	0.098445
38	Copy	298	0.098116
39	Author	297	0.097786

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 944855

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	215304	22.786988
1	MODIFIES	215304	22.786988
2	UPDATES	139782	14.794016
3	CREATES	52725	5.580221
4	INVOKES	37730	3.993205
5	DELETES	29815	3.155511
6	COMMITTED	23168	2.452016
7	DEPENDS_ON	22889	2.422488
8	OF_TYPE	22204	2.349990
9	DECLARES	21680	2.294532
10	OF_RAW_TYPE	17460	1.847903
11	HAS	14602	1.545422
12	HAS_PARENT	14170	1.499701
13	RETURNS	13073	1.383599
14	HAS_COMMIT	11584	1.226008
15	READS	9673	1.023755
16	HAS_ACTUAL_TYPE_ARGUMENT	8445	0.893788
17	HAS_FILE	8331	0.881723
18	RENAMES	7018	0.742759
19	OF_GENERIC_TYPE	6036	0.638828
20	RESOLVES_TO	4279	0.452874
21	SIMILAR	4130	0.437104
22	WRITES	4042	0.427791
23	CONTAINS	4005	0.423875
24	HAS_NEW_NAME	3662	0.387573
25	RETURNS_GENERIC	3616	0.382704
26	CHANGED_TOGETHER_WITH	3100	0.328093
27	ANNOTATED_BY	2943	0.311476
28	REQUIRES	2267	0.239931
29	HAS_FIRST_CHILD	2162	0.228818

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

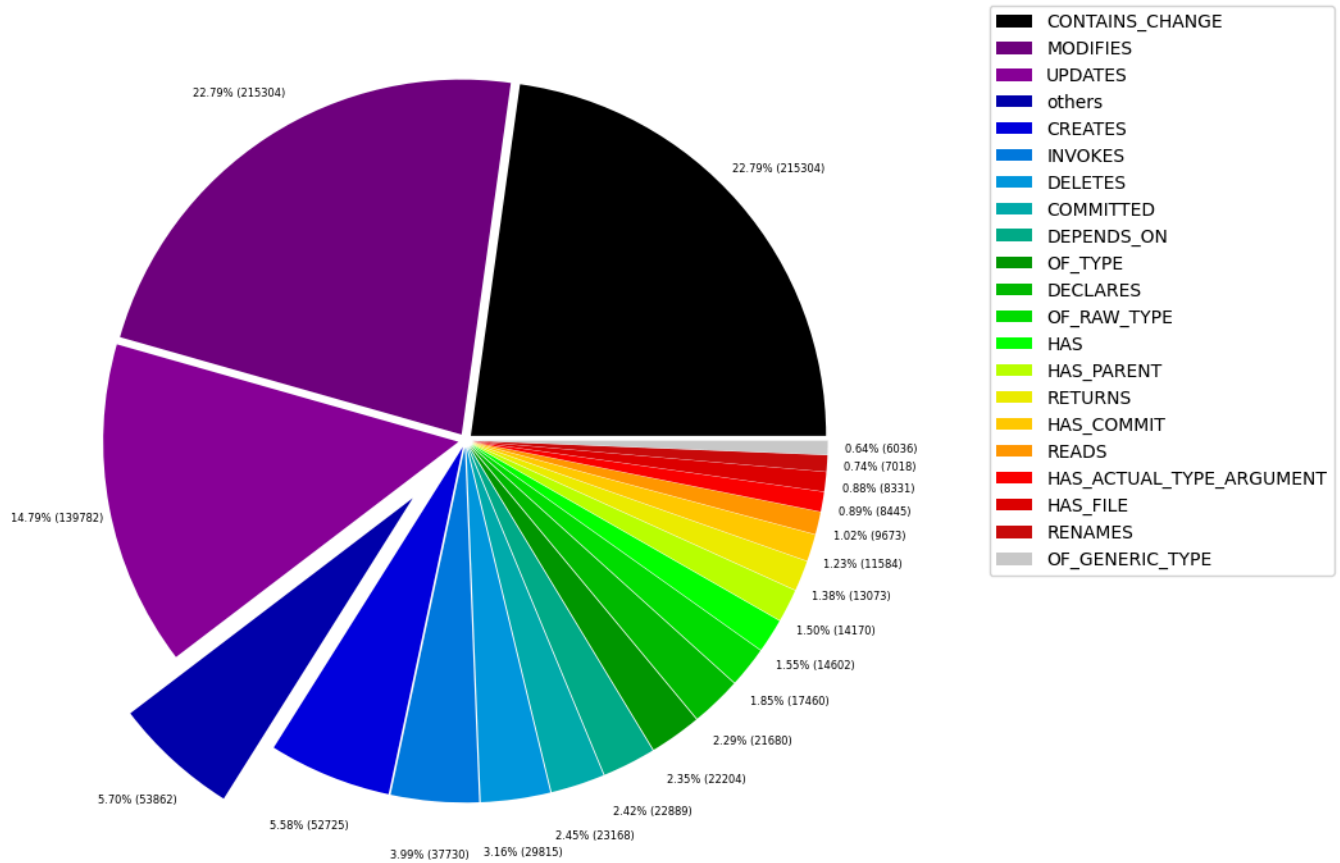


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

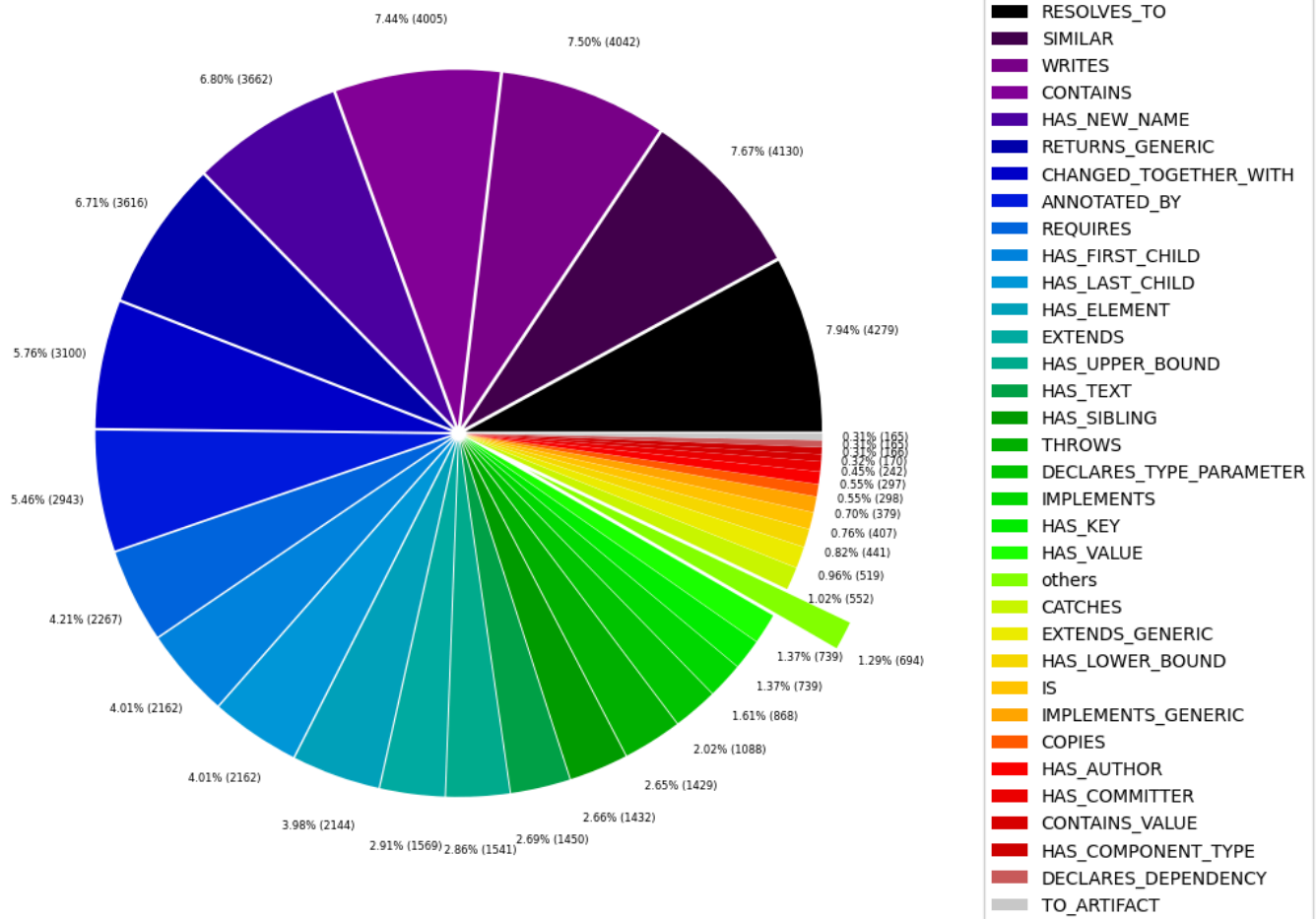
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000106
1	HAS_BRANCH	1	0.000106
2	HAS_HEAD	2	0.000212
3	THROWS_GENERIC	5	0.000529
4	EXCLUDES	5	0.000529
5	DESCRIBES	9	0.000953
6	HAS_ROOT_ELEMENT	11	0.001164
7	HAS_GOAL	16	0.001693
8	HAS_EXECUTION	16	0.001693
9	OF_NAMESPACE	18	0.001905
10	HAS_ATTRIBUTE	18	0.001905
11	INCLUDES_CONCEPT	19	0.002011
12	USES_PLUGIN	21	0.002223
13	IS_ARTIFACT	21	0.002223
14	HAS_CONFIGURATION	21	0.002223
15	REQUIRES_TYPE_PARAMETER	24	0.002540
16	REQUIRES_CONCEPT	28	0.002963
17	DECLARES_NAMESPACE	36	0.003810
18	HAS_DEFAULT	39	0.004128
19	COPY_OF	119	0.012595
20	HAS_TAG	132	0.013970
21	ON_COMMIT	132	0.013970
22	DECLARES_DEPENDENCY	165	0.017463
23	TO_ARTIFACT	165	0.017463
24	HAS_COMPONENT_TYPE	166	0.017569
25	CONTAINS_VALUE	170	0.017992
26	HAS_COMMITTER	242	0.025612
27	HAS_AUTHOR	297	0.031433
28	COPIES	298	0.031539
29	IMPLEMENTS_GENERIC	379	0.040112

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	139782	11584	
1	[Git, Update, Change]	MODIFIES	[File, Git]	139782	139782	
2	[Git, Update, Change]	UPDATES	[File, Git]	139782	139782	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	45409	11584	
4	[Git, Change, Create]	MODIFIES	[File, Git]	45409	45409	
5	[Git, Change, Create]	CREATES	[File, Git]	45409	45409	
6	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	23198	13687	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22797	11584	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22797	22797	
9	[Git, Change, Delete]	DELETES	[File, Git]	22797	22797	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	14161	11584	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11584	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11584	242	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	11584	297	
14	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8706	13687	
15	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8599	13687	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8331	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11584	
18	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6235	13447	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8331	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3488	7304	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3144	7316	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7316	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2709	13447	
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2577	2159	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2477	7316	

Graph Density

```
total_number_of_nodes (vertices): 303723
total_number_of_relationships (edges): 944855
-> total directed graph density: 1.0242624077983548e-05
-> total directed graph density in percent: 0.001024262407798355
```